

MODEL QUESTION PAPER1
S6 INSTRUMENTATION ENGINEERING
SUBJECT: MEMS AND NANO SENSORS

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	Cognitive Level
1	What is MEMS?	M1.01	R
2	List any two types of chemical sensors?	M1.02	U
3	What is mean by a substrate in microelectronics?	M2.01	U
4	What is called a lattice?	M2.02	U
5	What is the method used for depositing polycrystalline silicon on to a silicon wafer?	M2.03	U
6	List any two synthesized crystals which can be used as a piezo electric crystal?	M2.04	U
7	What is meant by dry etching?	M3.02	U
8	What is a base plate?	M3.04	U
9	What are Carbon Nanotubes?	M4.04	U

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	What are the major components of a microfluidics system?	M1.04	R
2.	Draw the schematic diagram of a micro valve.	M1.03	U
3.	Why the single-crystal silicon is the most widely used substrate materialfor MEMS and microsystems –give any three reasons?	M2.04	U
4.	Draw the schematic diagram of a piezoelectric transducer in a beam-accelerometer.	M2.04	U
5.	Explain dopant-controlled etch stop.	M3.03	U
6.	What is meant by stiction?	M3.02	U
7.	List the basic requirements of photoresist material.	M4.01	U
8.	What are the types of nanoscale sensors? Give examples for each.	M4.03	U
9.	Explain electrochemical probe technique.	M4.04	U
10.	Draw the schematic diagram of a glucose electrode.	M4.01	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Explain the components of a microsystem with block diagram.	M1.01	U
	OR		
IV	Explain pressure sensors with neat sketches.	M1.03	U
V	Explain the crystal structure of silicon.	M2.02	U
	OR		
VI	Write a short note on silicon piezoresistors.	M2.03	U
VII	Explain with neat sketches – deep reactive ion etching.	M2.02	U
	OR		
VIII	Explain surface micromachining process.	M3.01	U
IX	Explain nano scale biosensors for DNA sensing	M4.02	U
	OR		
X	Explain probe type sensing.	M4.04	U
XI	With necessary figure explain the working principle of chemical sensors.	M1.03	U
	OR		
XII	With necessary figure explain the operation microgrippers.	M1.04	U
XIII	Write a short note on quartz as a material for manufacturing MEMS.	M2.03	U
	OR		
XIV	List the applications of using polymers for manufacturing MEMS and microsystems	M2.04	U

Question Wise Analysis

Subject: **MEMS AND NANO SENSORS** – Question -1

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.01	R	1	2
I.2	M1.02	U	1	2
I.3	M2.01	U	1	2
I.4	M2.02	U	1	2
I.5	M2.03	U	1	2

I.6	M2.04	U	1	2
I.7	M3.02	U	1	2
I.8	M3.04	U	1	2
I.9	M4.04	U	1	2
II.1	M1.04	R	3	8
II.2	M1.03	U	3	8
II.3	M2.04	U	3	8
II.4	M2.04	U	3	8
II.5	M3.03	U	3	8
II.6	M3.02	U	3	8
II.7	M4.01	U	3	8
II.8	M4.03	U	3	8
II.9	M4.04	U	3	8
II.10	M4.01	U	3	8
III	M1.01	U	7	16
IV	M1.03	U	7	16
V	M2.02	U	7	16
VI	M2.03	U	7	16
VII	M2.02	U	7	16
VIII	M3.01	U	7	16
IX	M4.02	U	7	16
X	M4.04	U	7	16
XI	M1.03	U	7	16
XII	M1.04	U	7	16
XIII	M2.03	U	7	16
XIV	M2.04	U	7	16

Subject: **MEMS AND NANO SENSORS –QP-1**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	31 ± 6	2	2	2	6	4	28	8	36
2	16	33 ± 6	4	4	2	6	4	28	10	38
3	14	29 ± 6	2	2	3	9	2	14	7	25
4	15	30 ± 6	1	1	3	9	2	14	6	24
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	2	1.5
Understanding	121	98.5
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

MODEL QUESTION PAPER 2
S6 INSTRUMENTATION ENGINEERING
SUBJECT: MEMS AND NANO SENSORS

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	What is the range of size of components in MEMS?	M1.01	R
2	What is the principal application of an acoustic wave sensor?	M1.02	U
3	List any two materials used for manufacturing wafers.	M2.01	U
4	Which is the most widely used substrate material for MEMS and microsystems?	M2.02	U
5	List any two silicon compounds, often used in microsystems.	M2.03	U
6	List any two natural crystals which can be used as a piezo electric crystal?	M2.04	U
7	What are the two popular techniques used in etch stop?	M3.02	U
8	What is PMMA?	M3.04	R
9	How carbon nanotubes are classified?	M4.04	U

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	1. List any three advantages of microfluidics.	M1.04	U
2.	Draw the schematic diagram of a micro gripper.	M1.03	U
3.	List the popular polymer materials used for manufacturing MEMS and microsystems.	M2.04	U
4.	What are the advantages of gallium arsenide over silicon for considering it as a manufacturing material for microsystems?	M2.04	U
5.	What is meant by bulk micromanufacturing?	M3.03	U
6.	Explain electrochemical etch stop.	M3.02	U
7.	List any three advantages of dry etching over wet etching.	M4.01	U
8.	What are nanosensors? Give examples.	M4.03	U
9.	What is the use of electron microscopy in manufacturing nanosensors?	M4.04	U
10.	What is meant by nano scale Lithography?	M4.01	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Explain the operation of micromotor with neat sketches.	M1.01	U
	OR		
IV	Compare microelectronics and microsystems.	M1.03	U
V	What are the advantages of using silicon for manufacturing microsystems?	M2.02	U
	OR		
VI	Write a short note on silicon dioxide and silicon carbide.	M2.03	U
VII	Explain isotropic and anisotropic etching in detail.	M3.02	U
	OR		
VIII	Explain major fabrication steps in LIGA process.	M3.01	U
IX	Write short notes on a. Electrochemical probe and b. Spectroscopic technique.	M4.02	U
X	OR		
	Explain nano scale biosensors for DNA sensing with schematic.	M4.04	U
XI	Explain microaccelerometer with a schematic structure.	M1.03	U
	OR		
XII	Write a short note on microfluidics.	M1.04	U
XIII	Explain the process of polysilicon deposit on silicon substrate with a schematic.	M2.03	U
	OR		
XIV	What is LB film? List its applications in MEMS.	M2.04	U

Question Wise Analysis

Subject: **MEMS AND NANO SENSORS**– Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.01	R	1	2
I.2	M1.02	U	1	2
I.3	M2.01	U	1	2
I.4	M2.02	U	1	2
I.5	M2.03	U	1	2
I.6	M2.04	U	1	2
I.7	M3.02	U	1	2
I.8	M3.04	R	1	2
I.9	M4.04	U	1	2
II.1	M1.04	U	3	8
II.2	M1.03	U	3	8
II.3	M2.04	U	3	8
II.4	M2.04	U	3	8
II.5	M3.03	U	3	8
II.6	M3.02	U	3	8
II.7	M4.01	U	3	8
II.8	M4.03	U	3	8
II.9	M4.04	U	3	8
II.10	M4.01	U	3	8
III	M1.01	U	7	16
IV	M1.03	U	7	16
V	M2.02	U	7	16
VI	M2.03	U	7	16
VII	M3.02	U	7	16
VIII	M3.01	U	7	16
IX	M4.02	U	7	16
X	M4.04	U	7	16
XI	M1.03	U	7	16
XII	M1.04	U	7	16

XIII	M2.03	U	7	16
XIV	M2.04	U	7	16

Subject: **MEMS AND NANO SENSORS –QP 2**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	31 ± 6	2	2	2	6	4	28	8	36
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3	14	29 ± 6	2	2	3	9	2	14	7	25
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Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	2	1.5
Understanding	123	98.5
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100